Ben Chung

236-562-7866 | ben9354@gmail.com | Linkedin | Github | Website

Education

University of British Columbia Expected Grad: Nov 2027
Bachelor of Applied Science – Electrical and Computer Engineering

Experience

Junior Electrical Designer September 2025 – May 2026
Smith + Andersen

- Designed SLDs, power, and lighting layouts for tenant buildings using AutoCAD MEP, Revit, and Visual Lighting.
- Wrote EPRs to quantify electrifying existing equipment and implement EV charging. Determined electrical demand, spare capacity, and optimized power efficiency.
- Performed calculations to size feeders, circuit breakers, mech equipment, and panels according to BCBC and CEC.

Airside Project Management January 2025 – August 2025
Vancouver Airport Authority (YVR)

- Supported YVR's $170M North Runway Program by assisting with technical coordination, project documentation, and stakeholder communication across paving, drainage, and electrical work scopes.
- Created a master project tracker based on P6 schedule to determine earned value of completed work and modelled progress rates to extrapolate forecasted project completion dates.
- Created a manual to familiarize new users with YVR's Airfield Lighting System and Power Distribution System with adherence to TP312 aerodrome compliance standards.

Projects

Self-Balancing Robot (2016 Hoverboard) | *C++, Python, Control Systems, Electronics, Embedded Programming*

- Developed embedded C++ control software using a cascaded PID architecture, where the outer position PID adjusts target angle while the inner angle PID loop controls motor PWM output.
- Fine-tuned gyroscope and accelerometer filter values, PID gains, and encoder & motor response characteristics for precise robot balance and control.
- Integrated BLE connection with a Flutter mobile app to control robot movements.

iPod Machine with Volume Indicator | *SystemVerilog, Signaltap, Embedded Programming*

- Designed a keyboard-input FSM to control flash memory writes. Designed a separate FSM to synchronize address controller and flash memory reads.
- Programmed an interrupt routine to sample and real-time filter audio data from flash memory to visualize audio strength using an embedded PicoBlaze microcontroller.
- Verified system functionality using waveform simulations and SignalTap.

Omegle Chat Room | *Python, RTOS, TCP/IP*

- Built a multi-client and server-hosting TCP chat application using Python sockets, enabling clients and servers to exchange messages in real time.
- Implemented disconnect detection, status broadcasts, socket cleanup, and mutex lock to preserve message order and prevent race conditions.

RC4 Brute-Force Message Decryptor | *SystemVerilog, Signaltap, FPGA*

- Designed a 16-state FSM decryptor with pipelined reads and a start-finish controller communication protocol.
- Integrated RAM and ROM modules to read and write key scheduling, ciphertext storage, and decrypted output writes.
- Parallelized the brute-force key search across four processing cores to accelerate search throughput.

Technical Skills

Programming/HDL Languages: Python, C/C++, Java, SystemVerilog, Assembly
Proficent Programs: LTSpice, Matlab, Visual Studio, Git, AutoCAD, Revit, Quartus, SignalTap, MS365, Bluebeam
Awards: UBC Centennial Entrance Award, BC Achievement Scholarship
Hobbies: Boxing, Hiking, Volleyball